

BRYAN TANG

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Education

The University of Texas at Austin

August 2023 – May 2027

Bachelor of Science in Computer Science and Mathematics Double Major
Minor in Business

Austin, Texas

GPA: 3.6

Relevant Coursework

- Data Structures
- Operating Systems
- Computer Architecture
- Algorithms
- Artificial Intelligence
- Machine Learning
- Linear Algebra
- Differential Equations

Experience

Capital One Financial Company

June 2026 – August 2026

Incoming Software Engineer Intern

Richmond, Virginia

- Engineered a full-stack SFTP payables ingestion pipeline (**Go** APIs, **React** frontend, **PostgreSQL** backend) to automate downstream payment processing for commercial banking clients within Capital One's BankTech division.

Hewlett Packard Enterprise

May 2025 – May 2026

Software Engineer Intern

Spring, TX

- Wrote a **Cryptography Bill of Materials (CBOM)** generator that scans **Docker** containers with **Syft** and fuzzy string matching to inventory cryptographic and flag quantum-vulnerable algorithms for post-quantum migration.
- Developed an open-source CLI tool in **Go** to query server information via the **Redfish API**, automating firmware vulnerability detection and improving inspection efficiency by 300% over manual workflows.
- Engineered 7 **Cobra** CLI commands to automate auditing and validation pipelines, reducing auditing time by 40%.
- Architected a migration from a JSON datastore to an optimized **SQLite** database with advanced indexing strategies, reducing firmware record lookup latency by 65% and improving scalability for enterprise-level data workloads.

Applied Research Laboratories

January 2025 – May 2025

Software Engineer Intern

Austin, TX

- Built a **Java Swing** application with asynchronous, fault-tolerant data fetching and JSON parsing to visualize real-time sensor data via **REST API**, reducing data retrieval latency by 50%.

PCOnline

May 2024 – August 2024

Software Engineer Intern

Newark, DE

- Designed a full-stack **Python** application with a **Tkinter** GUI to automate label generation for 6000+ returned products, scaling warehouse reconciliation processes and improving tracking accuracy.
- Engineered automated UI testers using **Selenium** for inventory management tools, decreasing testing time by 90%.

Projects

POSIX Shell Implementation | C, C++

- Created a **POSIX**-compliant shell that supports 30+ commands, scripting, redirection, pipelining, and variables.
- Implemented a custom tokenizer and abstract syntax tree to parse logical operators and enable short-circuiting.
- Implemented background process support using `fork()` and `exec()` **system calls**, enabling concurrent process execution and improving responsiveness under load.

AI Firmware Vulnerability Triage Platform | Python, Go, FastAPI, Claude (LLM), scikit-learn

- Built a three-stage AI pipeline that auto-triages server firmware vulnerabilities: **RAG** semantic search (sentence-transformer **embeddings**) over a CVE corpus, a **gradient-boosted ML** exploitability model, and a **Claude LLM agent** using **tool calling** to rank findings and draft remediation.
- Served context-aware **P0–P4** triage reports through a **FastAPI/SQLite** backend and **Go/Cobra CLI**, with an offline RAG+ML fallback, a 36-test suite, and an evaluation harness.

Operating System | C, C++, x86

- Built a minimal **Unix-like** operating system kernel in **C/C++** and **x86** assembly.
- Implemented core OS components including process management, **ELF** binary loading, **system call** handling, memory management (**VMM**), synchronization primitives (semaphores, barriers), and basic file system support (**ext2**).

Technical Skills

Languages: Python, Java, C, C++, HTML/CSS, Go, SQL, JavaScript, Bash

Technologies: Linux, Git, Docker, Kubernetes, REST API, Redfish API, AWS, PostgreSQL, SQLite

Concepts: Operating Systems, Computer Architecture, Systems Engineering, Software Engineering, Automation/Scripting

Work Eligibility: Eligible to work in the U.S. with no restrictions